

Industrial Augmented Reality – Concept for an AR-AI-Station for Component Recovery from Washing Machines for Circular Economy

Mario Lorenz^{1abc}Martin Okoniewski^{2a}Arwa Own^{3a}Sebastian Knopp^{4a}

^aChemnitz University of Technology, Professorship for Production Systems and Processes; ^bUniversity Hospital Leipzig, Department of Orthopaedics, Trauma and Plastic Surgery; ^cMedical University of Graz, Division of Macroscopic and Clinical Anatomy

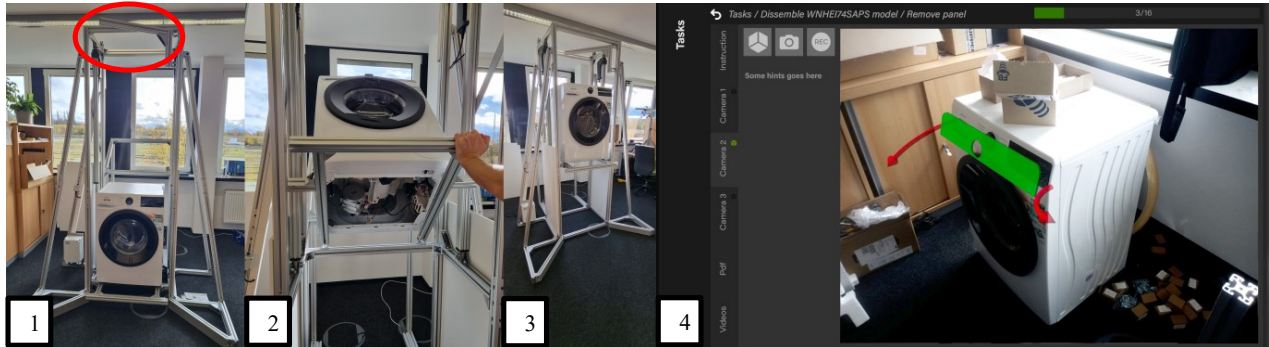


Figure 1: AR-Station for component recovery of Washing Machines in down position (1), up position (3) and tilted in up position (2). Movable arm for water tub recovery in red (1). UI-concept of AR-disassembly support system (4).

ABSTRACT

Using Augmented Reality (AR) in the production of home appliances is an understudied field, even more its use in circular economy. Further, also the application of Artificial Intelligence (AI) for validating that an AR instructed task was actually carried out correctly is an underexplored area. In this work we present the novel concept for an AR disassembly station to recover parts from returned washing machines in a circular economy approach, validating that the disassembly is carried out correctly using a YOLO-based AI model. A last challenge, tackled in this work resulting from the circular economy background, is the adaptability of the AR instructions that need to change depending on the varying conditions of the recovered parts in units of the same model.

Keywords: Augmented Reality, Artificial Intelligence, Circular Economy, Disassembly, Validation.

Index Terms: Human-centered computing → Mixed / augmented reality; Computing methodologies → Artificial intelligence

1 INTRODUCTION

In numerous industrial settings, Augmented Reality (AR) is applied to boost design, maintenance, and assembly activities. The transition from sequential to a circular economy presents new challenges for AR systems, particularly in disassembly. Disassembly is crucial for recovering components from products for reuse in a circular economy. Unlike during assembly, where standardized products are produced, the disassembly of used products involves deal-

ing with their varying conditions. The standardized products become individual products during the disassembly. The EU-funded DiCiM project (www.dicimproject.eu) is focused on developing a support system for disassembling washing machines, where each unit requires unique handling. For instance, one washing machine may have a defective motor needing disposal, while another may have a reusable motor, and a third may require refurbishment. Thus, an AR system must provide tailored instructions for each machine. Additionally, ensuring the correct execution of tasks via AR instructions poses a challenge. Our monitor-based AR disassembly system intends to incorporate an Artificial Intelligence (AI)-based validation mechanism to verify task completion accuracy. We present our concept for an AR-AI station for component recovery from washing machines, which shows potential applicability in other domains.

2 STATE OF THE ART

AR has been utilized for a long time for assembling products [1]. However, for disassembly the literature is much sparser, with recent applications for batteries [2] and teleoperated disassembly [3]. So far, no AR application supporting the disassembly for of washing machine in particular nor home appliances in general have been reported. Here, AR is mainly applied for the interaction with the home appliance [4] or marketing applications [5].

The integration of AI and AR has manifested in numerous applications in recent years [6], although primarily aimed at improving AR technology itself rather than facilitating task validation. To date, a limited number of studies have addressed the automated validation of tasks instructed by AR. Manuri et al. [7] and Alves et al. [8] employ a computer vision-based methodology for the validation of tasks related to the maintenance of a laser calibrator and the assembly of LEGO models, respectively. In contrast, only Rheault et al. [9] have applied AI techniques to navigate users through a cooking task utilizing AR.

3 PROCESS OF WASHING MACHINE DISASSEMBLY

The general disassembly of a washing machine starts with removing the top and disassembling of the accessible parts. Next, the back

¹mario.lorenz@mb.tu-chemnitz.de.

²martin.okoniewski@mb.tu-chemnitz.de.

³arwa-yaseen-ahmed.own@mb.tu-chemnitz.de.

⁴sebastian.knopp@mb.tu-chemnitz.de.

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is removed for disassembling the accessible parts from there. After that the washing machine is tilted approx. 45° to the back to remove the parts accessible from below. Lastly, it is put back on its feet again and the parts accessible from the front are recovered before the water tub is lifted out of the washing machine using a wrench. It is not possible to tilt the washing machine by 90° during disassembly, as the washing unit would be damaged.

4 AR-STATION CONCEPT

In order to support the washing machine disassembly with AR we decided to build a station (1.70 m width, 2.30 m height, 1.50 m depth) that could lift the washing machine by 1.00 m using two LINAK electrical cylinders (1.25 m length) and tilt it in this position by 45° (see Figure 1). This allows a more ergonomic disassembly compared to the current process where the washing machine is grounded. For recovering the washing unit, a wrench is put on top of the AR-station and the wire is guided over a movable arm, which allows to pan the recovered washing unit out of the AR-station (see Figure 1, 1). On all four sides, the AR-station is equipped with structure hindering its toppling. Further, we decided to use a touchscreen-monitor-based AR setup with multiple Intel RealSense cameras (not yet attached in Figure 1). A projector-based approach was ruled out due to the narrow working space resulting in many self-occlusions. Head mounted displays were not considered due to the narrow working area, the low battery capacities, the high strain on the eyes resulting from the vergence-accommodation conflict in case of a video-see-through display respectively the small field of view in case of an optical-see-through display.

5 AR-USER INTERACTION CONCEPT

The AR-application is interactable via a touchscreen monitor and is mainly providing a list of available instructions for different washing machines. The disassembly instruction is presented as step-by-step information about the disassembly of the washing machine and the sorting of the components that should be recovered. The instructions is conveyed to the worker using 3D-model overlays, texts, images and videos depending on the most suitable option for a step (see Figure 1). Here, the worker can choose between the different camera perspectives, and is able to take a photo or video in case they detect an issue that needs to be forwarded to the refurbishment managing engineers. For this reason, the AR-application also includes a notification function, that allows to send such a report. When the disassembly starts, a predictive condition monitoring system provides information on conditions of the parts that should be recovered to the AR application. This causes an adaption of the instructions, e.g. the motor of the current washing machine if found to be reusable, the worker is instructed to put it in the reusable box. For the next washing machine of the same type the motor might be broken, so at the same point during the disassembly the worker is instructed to put it in the recycling box.

6 AI-VALIDATION CONCEPT

The concept of AI supported validation of the disassembly tasks is to utilize the YOLO [10] model for detecting the components of the washing machine. The YOLO model is trained with images from the video stream of the intel RealSense cameras. At the beginning of each disassembly step the AR model checks in the background if the part to be recovered is present. When moving to the next step the AI model checks the part isn't detected anymore, hence assuming it was recovered.

7 FUTURE WORK

The presented concept of the AR-AI-station for component recovery from washing machines for circular economy application is in a development state where the structure of the station is mostly built

and needs to be completed with the AR hardware next. Further, the AR application and AI model will need to be developed and merged to enable a validation of the presented concepts. Once it is complete the applications user experience, user satisfaction, ease of use and time performance should be evaluated. Further a transfer of the AI-validation approach of AR instructed tasks to other household appliances and industrial sectors should be pursuit.

8 CONCLUSION

AR is rarely used for manufacturing of home appliances in general. In particular, AR use for washing machines is under studied. We presented the early concept stage of an AR application that can support the component recovery from returned units. We further introduce AI to the overall concept which will be used to detect if an instructed task was carried out correctly, a further understudied research field.

ACKNOWLEDGEMENTS

The authors acknowledge financial support from the European Union's Horizon Europe research and innovation programme under the grant agreement No. 101091536 (DiCiM project).



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